

Rethinking the Fabrication of Interactive Devices Through Biomaterials: Towards New Paradigms of Making and Materiality in HCI

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Abstract

Human-Computer Interaction (HCI) has increasingly turned to biodegradable or bio-based materials as a response to ecological concerns. They are often used to replicate the functions of synthetic counterparts, leaving dominant fabrication paradigms unquestioned. This limits the field's ability to engage with the temporal and responsive qualities of biomaterials. My research explores how biomaterials, both bio-based (bioplastics) and living (bacterial cellulose), can transform not only what interactive systems are made of, but also how they are designed, fabricated, and understood. By leveraging their physical, temporal, and ecological properties, I propose new workflows and develop fabrication frameworks that place material transformation at the center of interaction design, reframing making as a temporal, co-constitutive practice. This shift invites new computational imaginaries, where material transformation itself, through growth, decay, or regeneration, becomes a channel for sensing, actuation, or control.

CCS Concepts

• **Human-centered computing** → **Interaction devices**; • **Hardware** → **Emerging interfaces**; **Bio-embedded electronics**.

Keywords

Bio-HCI, Biofabrication, Biomaterials, Living Systems, Fabrication, Sustainability, Making, Materiality

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1 Introduction

Advances in material synthesis and mass fabrication have made computing hardware a global, performant commodity, i.e., widely available, standardized, and cost-effective, but also fueled a culture of disposability. Although many electronic components are technically recyclable [20], most devices are neither designed for repair nor reuse, resulting in premature disposal and growing e-waste challenges [8, 34]. Human-Computer Interaction (HCI) community has responded with efforts to reduce energy consumption, extend device lifespans, and promote circular design [20]. These strategies largely operate within fabrication paradigms that assume materials are relatively stable over time, standardized, and controllable. Such assumptions limit engagement with emerging materials whose properties are intentionally transient, i.e., designed not for permanence, but for transformation, reintegration, or decay, which are core qualities of sustainable design.

Materiality in HCI is not just a medium, but a mode of inquiry and a method [10, 13, 33]. As Material-Driven Design gains traction, biomaterials (i.e. bio-based, and biodegradable materials often grown or transformed through biological processes) have emerged as both practical alternatives and conceptual provocations. Derived from living or natural processes, they bring new affordances such as growth, transformation, responsiveness, and decay into the design space, challenging existing workflows and core values like control, uniformity, and efficiency. Rather than being fixed endpoints, these materials are approached as processes that unfold over time [1, 5, 11, 16, 17, 24]. Thus, growth [4, 11, 24], decay [3, 31], multi-species entanglement [25, 36], and responsiveness [6, 14, 26, 28, 35] are treated not as problems to be engineered away, but as design resources. Designers in this space increasingly engage with care, attunement, and uncertainty, shifting how interaction is prototyped and understood and reconfiguring design knowledge as procedural, situated, and ecological.

Despite their promise, HCI methods, tools, and workflows to support full integration of biomaterials are yet limited. As a result, they are often dismissed as impractical or retrofitted into conventional design processes.

My research explores how the inherent capabilities of biomaterials and their biological processes can be leveraged to develop new fabrication frameworks and interaction paradigms for HCI. To do

this, I develop hybrid fabrication methods that combine material engineering with generative, craft-based techniques, thus bridging structured reproducibility with the dynamic behaviors of biological matter. My approach is grounded in experimental making and material exploration, resulting in functional prototypes such as soft sensors, shape-changing devices, and biohybrid systems. I evaluate these systems through iterative testing and material characterization, and qualitative research, including expert interviews, as well as critical reflection on the broader infrastructural, methodological, and perceptual shifts they introduce to HCI. Together, these investigations aim to establish a foundation for new fabrication paradigms that position biomaterials as co-designers in sustainable interaction design. In the following sections, I summarize three research frameworks and their contributions to date, along with the methodological insights that emerged through their development.

2 Fabricating with Biomaterials

My research began with a specific question: could biomaterials replicate the functions of conventional interactive materials while offering an approach that is accessible for a broad audience? I focused on biopolymers, specifically bioplastics, a bio-based and often biodegradable alternative to conventional plastics. DIY bioplastics (e.g. made from gelatin, agar, starch, or alginate) have gained popularity as renewable materials, supported by open-source recipes and databases [9, 22]. Recent work functionalized them with carbon additives for conductivity and sensing [19, 27], but most approaches remain limited to lab environments due to material safety and processing complexity [15].

To bridge this gap, I investigated how bioplastics could support sensing, acting not only as substrates but also as conductors. This revealed key differences from synthetic substrates, particularly in their response to moisture, deformation, and electrical interfacing. Moving toward actuation, the challenge shifted: creating soft, responsive materials required a broader palette of tunable properties such as elasticity, water absorption or and low shrinkage.



Figure 1: Soft input and output devices made with carbon-infused bioplastics using the Interactive Bioplastics framework and DIY-friendly methods.

2.1 Interactive Bioplastics (ACM UIST '22)

The Interactive Bioplastics framework [16] emerged from this initial inquiry into sensing. It leverages the biodegradability of biomaterials and focuses on the replacement of traditional substrates and conductive materials with low-cost bio-based and bio-degradable alternatives. It introduces substrates and three carbon-infused material types (sheets, pastes, and foams) compatible with digital fabrication tools and conventional electronics. These materials support the creation of soft interactive devices (Fig. 1) such as on-body circuits (top), pressure sensors, self-contained computing devices and body-contacting electrodes for electromyography (bottom) that can be disassembled and re-molten. These examples illustrate how bioplastics can effectively replace traditional materials, maintaining functionality while significantly reducing environmental impact.



Figure 2: Functionalizable properties and processing techniques (top) used to design soft biopolymer actuators capable of volume, orientation and form change (bottom) in the SoftBioMorph framework. ©Madalina Nicolae

2.2 SoftBioMorph (ACM DIS '24)

Building on Interactive Bioplastics, the SoftBioMorph framework [24] was shaped by the realization that actuator design required more than elasticity: it required fine-tuning material parameters such as hydration, shrinkage, and stiffness. However, synthesis methods were scattered across fields like food engineering, and packaging, often buried in technical or lab-specific protocols. Therefore, SoftBioMorph leverages the customizability of biomaterials to enable dynamic behaviors in soft interactive systems. Focusing on soft biopolymer actuators, it introduces accessible techniques for synthesizing, assembling, and interfacing bioplastics into systems capable of three primitive types of shape change: volume, orientation, and form (Fig. 2 top). Five application cases (bottom) illustrate how bioplastics can be customized to achieve dynamic behaviors,

demonstrating their potential to support a broader range of output modalities and interaction opportunities. By offering accessible control levers over elasticity, thickness, hydration response, and drying dynamics, the framework highlights the versatility of bioplastics beyond static forms. It expands the expressive potential of sustainable materials and supports the broader integration of shape-changing behavior into ecological interaction design.

Together, these frameworks expand the scope of material experimentation in HCI, showing that biomaterials can be both functionally robust and environmentally aligned. While bioplastics may be sensitive to environmental conditions and mechanical stress, traits labeled as “limitations” in the Interactive Bioplastics framework, they also open possibilities for using material responsiveness as a cue for sensing, controlled transformation, or programmed degradation. The SoftBiMorph framework illustrates this potential by exploiting ionic sensitivity to alter material structure.

3 Growing Biohybrid Systems (ACM UIST '23)

While the previous frameworks made a step forward, they remained rooted in a logic of replication: applying biomaterials only after processing, i.e., once their biological activity had ceased. This led me to a deeper question: what new affordances arise when livingness, not just biodegradability, is treated as a core material property in interactive device fabrication? To explore this, I turned to bacterial cellulose, a living biomaterial that self-assembles during fermentation. Unlike static biopolymers such as bioplastics, its growth can be guided, interrupted, or merged with other materials, offering new possibilities for how devices take shape. While co-fabrication with living organisms has emerged as a promising paradigm in design and HCI [7], bacterial cellulose has largely been used to create passive forms through post-processing [18, 29, 30]. However, the potential of such materials for creating functional, responsive systems by embedding biological growth into the design process remained largely underexplored.

This led to the development of the Biohybrid Devices framework (ACM UIST '23), which reimagines how living biomaterials can be integrated into the fabrication of interactive devices by treating biological growth as an active, designable process rather than a post-production material state. The framework introduces lifecycle-centric fabrication techniques that embed electronics during three distinct phases: Growth, Stabilization, and Inanimate. This approach departs from traditional assembly-based paradigms by leveraging the temporal and self-organizing properties of living systems as part of the fabrication itself (Fig. 3). The framework supports a range of functional devices (Fig. 4) among which a game controller, illustrating how living materials can act as both structural and functional components, responsive not just in a mechanical sense, but in their evolving relationship to the environment.

While focused on bacterial cellulose, its principles are adaptable to other living organisms, laying the groundwork for broader integration of biological processes into HCI. This work challenges the boundaries between organism and object, and expands HCI's material and procedural vocabulary through the lens of livingness. Traits such as growth variability, moisture dependence, and slower production times are often seen as barriers to industrial reproducibility

and rapid prototyping. Yet, these same traits open opportunities for “co-designed” growth processes, where material development is shaped in dialogue with environmental conditions and embraces longer temporal cycles.

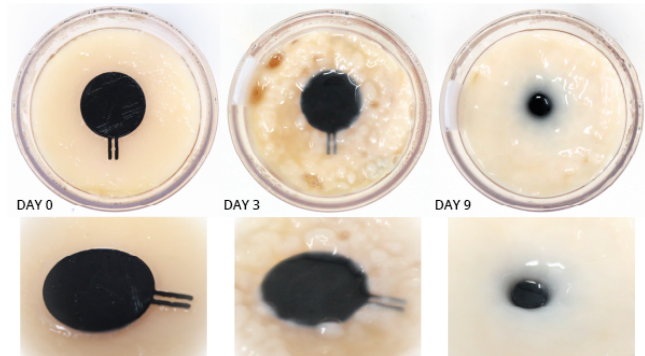


Figure 3: Photographic sequence of bacterial cellulose growing around a conductive 3D printed pad over several days, exemplifying *Grow around method*, one of the core techniques in the Biohybrid Devices framework. This allows electronic components to become structurally and functionally integrated as the material self-assembles. ©Madalina Nicolae & Vivien Roussel

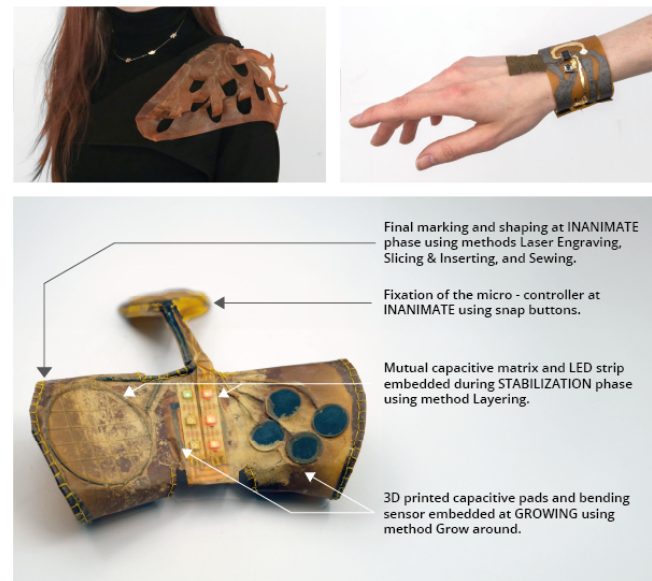


Figure 4: Three prototypes developed using the Biohybrid Devices framework: a wearable shape-changing accessory (top left), an interactive bracelet (top right), and a game controller annotated to illustrate the integration of interaction elements across all lifecycle phases—Growth, Stabilization, and Inanimate (bottom). These prototypes show how bacterial cellulose can support both structural and functional roles in interactive systems. ©Madalina Nicolae & Vivien Roussel

4 Reflecting on Biofabrication in HCI

Biofabrication has emerged in HCI as a way to rethink fabrication through biological processes, building on concepts such as Material-Driven Design [13] and living interfaces [23]. Prior research spans taxonomies and frameworks [12, 32], subjective first-person engagements with biomaterials [2, 25] and speculative accounts that expand the field's conceptual scope [23]. Yet, these contributions often lack generalizable insights into reproducibility or workflow integration, leaving them disconnected from the practical realities of working with biomaterials. Although workshops and expert interviews have begun surfacing frictions [3, 21, 31], a systematic account of biofabrication's infrastructural, epistemological, and material challenges is still missing.

Taking a step back, in each of the previous frameworks, traits related to speed, robustness, or environmental sensitivity were labeled as "limitations" following dominant expectations in HCI, rather than emerging from the actual design process. In practice, I approached these not as problems to resolve, but as design cues and entry points to reflect on deeper systemic tensions in how biofabrication is evaluated and practiced in HCI.

To explore these tensions more systematically, I conducted a qualitative study (currently in preparation for submission) based on 15 in-depth interviews with experts in HCI, biology, design, and industry. The study maps fabrication practices across HCI and related fields, clarifies conceptual ambiguities, and identifies systemic obstacles, shared tensions, and unspoken values that shape biofabrication workflows.

The analysis reveals four overarching challenges: (1) a misalignment between biofabrication's values, such as sustainability, relationality, and material temporality, and conventional metrics of speed, scale, and control, leading to difficulty in evaluating biofabrication within existing HCI benchmarks and its frequent dismissal as inefficient or impractical; (2) a mismatch of material expectations, where biomaterials are often treated as direct substitutes for industrial ones, overlooking their temporal behaviors, non-standard properties, and ecological affordances; (3) high entry barriers, as biofabrication, despite its DIY framing, requires care-intensive routines, extended timeframes, and tacit knowledge making workflows difficult to standardize, reproduce, or scale and (4) a lack of foundational data and computational models, which limits designers's abilities in simulating or predicting biological behaviors, making reproducibility and cross-domain collaboration and tooling difficult. Although familiar, these challenges are analyzed both technically and experientially, shaping how designers think, feel, and act within biofabrication workflows.

From these insights, the study outlines seven concrete research opportunities spanning new evaluation frameworks, interdisciplinary collaboration protocols, generative design tools, and infrastructures for knowledge sharing and care. Each addresses distinct but interconnected layers of a broader solution to the systemic challenges identified, forming a pathway toward more integrated, generative approaches to biofabrication.

Finally, the study reflects on broader implications, including public perception, cultural narratives, and ethical concerns, positioning biofabrication both as a technical endeavor and a paradigm shift in how materials, processes, and sustainability intersect in HCI.

Together with my collaborators, we hope this study contributes to a broader research agenda in biofabrication and bio-integrated design in HCI that redefines how we engage with making, materials, and sustainability.

5 Next Steps

I am currently completing the fourth year of my doctoral research, conducted jointly at Saarland University's Department of Computer Science, De Vinci Research Center at De Vinci Higher Education and the Institut Polytechnique de Paris, under the supervision of Prof. Jürgen Steimle, Dr. Marc Teyssier and Dr. Samuel Huron. I have published three peer-reviewed papers directly related to my dissertation, with a fourth currently in preparation. In parallel, I am also developing a new project, BioWeave, which tests a new trajectory I envision for bio-integrated systems.

Looking ahead, I see biofabrication not as a niche exploration or sustainable alternative, but as the foundation for a broader shift in how we engage with making in HCI. I believe that working with living materials invites us to reconsider what design is, i.e. what it produces, how it unfolds over time, and who or what participates in its realization. Rather than retrofitting biological matter into familiar artifacts, I'm increasingly drawn to designing with growth, treating biological transformation itself as a driver of form, structure, and interaction.

In my future work, I want to deepen this direction by focusing on materials that regenerate, thicken, or evolve, thus shaping the temporality and adaptability of interactive systems. Whereas morphogenesis is often understood as a developmental process toward a fixed form, I'm interested in morphing as a continuous, bidirectional process, where material transformation remains open, reversible, and responsive, i.e., an active, ongoing component of interaction. BioWeave, marks the beginning of this path, exploring how biohybrid systems might become living structures whose interactive properties change through growth. This direction raises a broader question: how might ongoing biological transformation act as a reconfigurable parameter for control or computation in interactive systems? With this, I aim to move beyond prototyping toward a biologically integrated design practice that expands HCI's material, conceptual, and ecological vocabulary, contributing shared tools and frameworks for materially interdependent and ecologically conscious infrastructures.

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